

Supplementary Material: An Empirical Analysis of Iterative Generation and Evaluation for Image Creativity

Yoshia Abe, Ryo Terajima, Tatsuya Daikoku and Yasuo Kuniyoshi

S1 Methods

S1.1 System Design

In this study, we constructed an image generation system that integrates automatic evaluation with iterative generation, as shown in Fig. 1 in the main paper. The system consists of three components: the Artist agent F_{Artist} , the Generator tool $F_{\text{Generator}}$, and the Critic agent F_{Critic} . The Artist agent and Critic agent are implemented using LLMs that can process not only textual input but also visual input. The Generator tool is an image generation model. Let the time step be t , the prompt text for image generation be p_t , the generated image be i_t , and the evaluation of the image be e_t . The system is initially given a theme s_{theme} . First, the Artist agent receives the theme text and outputs a prompt for image generation: $p_t = F_{\text{Artist}}(s_{\text{theme}})$. Then, the Generator tool receives the prompt and generates an image: $i_t = F_{\text{Generator}}(p_t)$. Lastly, the Critic agent receives the generated image and outputs the evaluation: $e_t = F_{\text{Critic}}(i_t)$. From the second loop onward, the Artist agent receives the previous prompt and evaluation in addition to the theme: $p_t = F_{\text{Artist}}(s_{\text{theme}}, p_{t-1}, e_{t-1})$.

We designed the following experimental setups:

- Generator Only (G): The theme is input into the Generator to produce an output.
- Artist + Generator (AG): The theme is given to the Artist, who creates a prompt for image generation. This prompt is input into the Generator to produce an output.
- Artist + Generator + Critic (1 loop) (AGC1): The system is executed with a single loop, terminating after the Artist and Generator respond to the Critic’s feedback on the initial generation.
- Artist + Generator + Critic (100 loops) (AGC100): The system is executed for 100 iterations, terminating after the Generator updates the output following the Critic’s 100th feedback.

The LLMs used in the Artist and Critic agents were either GPT-4o mini (version "gpt-4o-mini-2024-07-18") [5] or Claude 3.5 Haiku (version "claude-3-5-haiku-20241022") [1]. Hereafter, for simplicity, we refer to these models as GPT and Claude, respectively. Accordingly, a total of seven generation methods were prepared. To account for variability, each method was executed with 10 different random seeds. The Generator tool $F_{\text{Generator}}$ was implemented using Stable Diffusion [7]. We fixed the random seed in Stable Diffusion so that it operated deterministically: the same input always produced the same image. Similarly, we set temperature = 0 and top_p = 0 for the LLMs used in the Artist and Critic agents, but perfect deterministic behavior could not be fully guaranteed due to variability in server-side environments. We used 20 AI-generated themes covering everyday scenes, natural landscapes, objects, symbols, and abstract concepts, such as "City street corner", "Forest autumn leaves", and "Dreams in motion". The 20 themes were divided into two sets: 10 for Likert-scale evaluation and 10 for pairwise evaluation.

S1.2 Evaluation of Creativity

S1.2.1 Human Subjective Evaluation

We recruited participants online and asked them to evaluate the creativity of generated images. In addition to creativity, we collected data on various other aspects, including evaluations of novelty, surprise, valuableness, uniqueness, thought-provoking quality, artistry, beauty, preference, and thematic alignment, in order to examine the relationships among these factors. We prepared two evaluation methods. In Test-Human-1, participants were shown one image at a time and asked to rate it on the ten metrics on a 9-point Likert scale (1 to 9). In Test-Human-2, in order to compare two generation methods, participants were shown two sets of ten images, each set generated by a single method, and asked to choose which set was higher on each metric. In the human evaluation, we used counterbalanced random image subsets to reduce participant burden. A total of 513 participants (aged 20s-50s) took part; after excluding three participants with missing data, 510 participants were included in the analysis.

S1.2.2 LLM as a Judge

An example of employing LLMs for the automatic evaluation of generated outputs can be found in [4]. While the creativity of generated stories is evaluated in their study, we investigated creativity in the image modality. The LLM-based evaluation employed the same question items and response formats as in the human evaluation. In Test-Claude-1 and Test-GPT-1, the LLMs rated the creativity of individual images using the same 9-point Likert scale as in Test-Human-1. In Test-Claude-2 and Test-GPT-2, the LLMs were presented with two sets of ten images and were asked to select which set was more creative, as in Test-Human-2. In LLM-as-a-Judge setting, it is common to use a more powerful model than the one that generated the outputs to be evaluated. Therefore, we employed GPT-4o (version "gpt-4o-2024-08-06") [6] and Claude 3.7 Sonnet (version "claude-3-7-sonnet-20250219") [2] as judges.

S1.2.3 Measurement of Diversity

Based on cognitive science findings that identify divergent thinking as a key component of creativity [8], we designed a quantitative metric to evaluate the diversity of generated images as a supplementary analysis, following prior work [4]. To capture qualitative rather than pixel-level variations, we projected the generated images ($256 \times 256 \times 3$ pixels) into a lower-dimensional space and calculated the variance within that space. Specifically, image embeddings were extracted using a Vision Transformer pretrained on the ImageNet-21k dataset [3]. The embeddings were then projected into a two-dimensional space using Principal Component Analysis. Based on the variance of ten generated images produced with different random seeds in this two-dimensional space, we defined a diversity score as the sum of the variances along each principal component axis. In this analysis (Test-Diversity), we computed the diversity metric of each generation method across the 20 themes.

S2 Results

S2.1 Correlations between creativity and other metrics

We analyzed the results of Test-Human-1 to facilitate a quantitative comparison of creativity scores across the generated images. This test included 700 images generated from 10 themes, across 7 methods, and 10 seeds each. For each image, participants rated the 10 metrics (novelty, surprise, valuableness, uniqueness, thought-provoking quality, artistry, beauty, creativity, preference, and thematic alignment) on a 9-point Likert scale. Those scores for each image were calculated by averaging the ratings from multiple participants. Next, the scores for each method

and theme were calculated by averaging the scores over the 10 seeds. By further averaging these scores across the 10 themes, we obtained the final scores of the 7 generation methods. We computed the correlation coefficients between the creativity scores and the various evaluation metrics, as shown in Fig. S1. The results indicate that the creativity scores of generated images exhibit significant positive correlations with novelty, surprise, valuableness, uniqueness, thought-provoking quality, and artistry. In contrast, although beauty and preference show positive correlations with creativity, these correlations are not statistically significant, and thematic alignment exhibits a negative correlation.

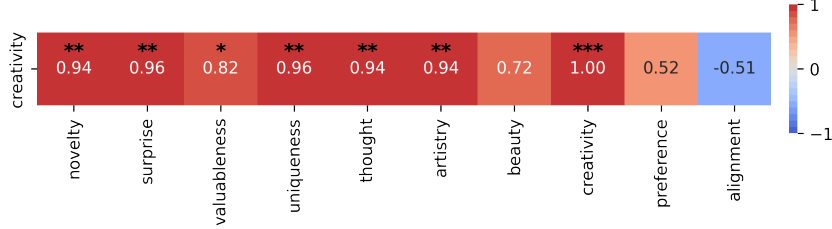


Figure S1: Pearson’s correlation matrix of evaluation metrics. Thought denotes thought-provoking quality, and alignment denotes thematic alignment. To account for multiple comparisons, p-values were adjusted using the Benjamini–Hochberg correction procedure to control the false discovery rate. Asterisks indicate statistical significance after the correction (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

S2.2 Change in diversity over iterations

To examine the effect of iterative refinement on diversity in greater detail, we conducted an additional analysis of two generation methods—AGC100-claude and AGC100-gpt—by calculating the diversity metric of the generated images across 10 seeds at every 10 loop iterations. The diversity trends averaged across all 20 themes are shown in Figure S2. From this figure, we observed that, for both the AGC100-claude and AGC100-gpt methods, diversity at step 1 was higher than at step 0. This supports the finding that performing a single loop involving the Critic (AGC1) results in higher diversity than using only the Artist and Generator (AG). On the other hand, we observed a difference between AGC100-claude and AGC100-gpt after step 10. The linear fit showed an increasing trend for AGC100-gpt and a decreasing trend for AGC100-claude.

S2.3 Correlations among evaluation protocols

We compared the rankings of the generation methods across evaluation protocols. In Test-Human-1, Test-Claude-1 and Test-GPT-1, a total of 700 images—generated from 10 themes, 7 methods, and 10 seeds—were evaluated using a scoring-based evaluation and the seven methods were ranked in terms of the mean score for each theme. In Test-Human-2, Test-Claude-2 and Test-GPT-2, a total of 700 images were evaluated using pairwise comparison and the seven methods were ranked in terms of the win counts for each theme. In Test-Diversity, the seven methods were ranked in terms of the diversity score for each theme. We then computed the mean rank for each generation method by averaging its theme-wise ranks across all themes, resulting in a seven-dimensional rank vector for each evaluation method. Spearman’s correlation coefficients were calculated between these rank vectors, as shown in Fig. S3. The correlation between Test-Claude-1 and the human tests indicates similarity of the Claude-based Likert-scale evaluation to human evaluation. Correlations between Likert-scale and pairwise tests should be interpreted with caution, because the two formats used different theme sets.

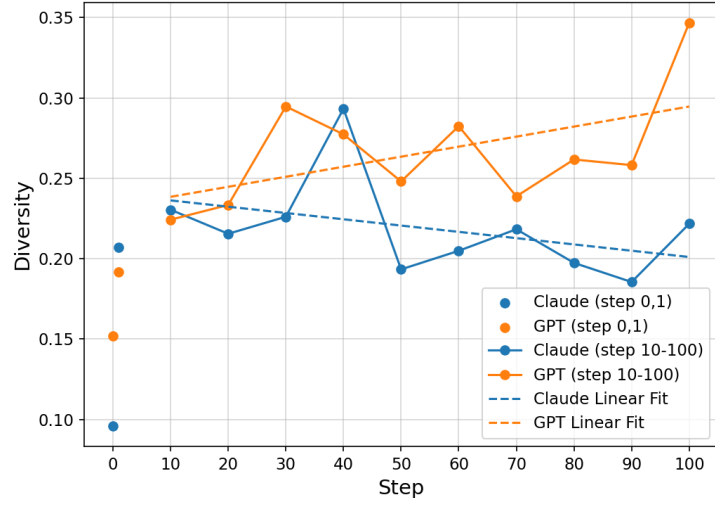


Figure S2: Mean diversity transition of AGC100-claude and AGC100-gpt across 20 themes. Steps 0 and 1 are plotted as individual points, while steps from 10 onward are plotted as a line graph at 10-step intervals. Dashed lines indicate linear regression lines fitted to the data from step 10 onward.

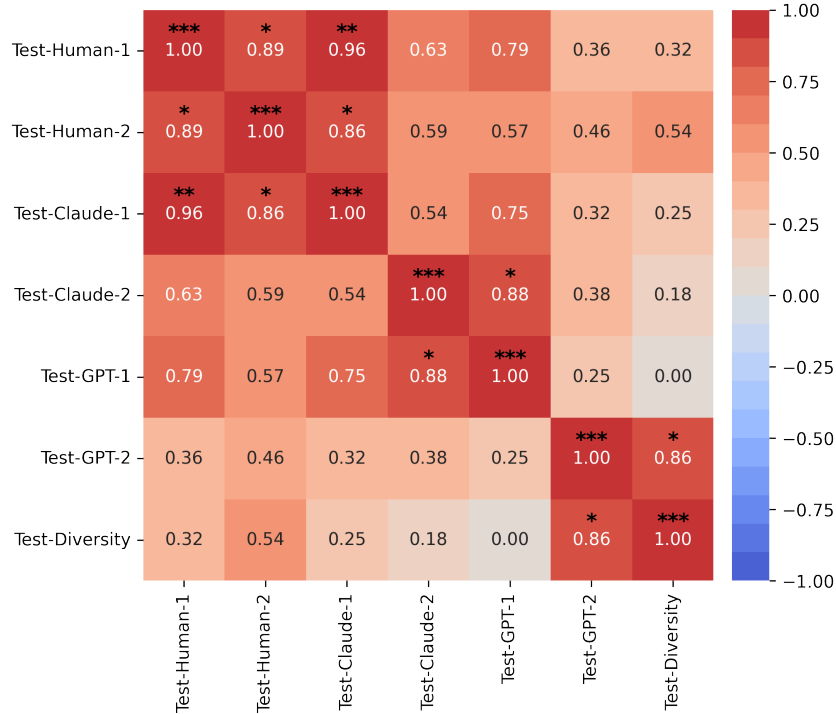


Figure S3: Spearman's correlation coefficients between different evaluation methods. To account for multiple comparisons, p-values were adjusted using the Benjamini–Hochberg correction procedure to control the false discovery rate. Asterisks indicate statistical significance after the correction (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

S3 Appendix

S3.1 Prompts

The following sentences are included in the system prompt for the Artist agent:

You are an expert in creating visual art using AI image generators. Your task is to generate an art piece based on the given theme. Create a prompt for an AI image generator that describes a non-photorealistic artistic image, specifying what is depicted and in what artistic style. Use descriptive, imaginative language that evokes a specific mood, theme, and visual style.

The following sentences are included in the system prompt for the Critic agent:

You are a critic specializing in evaluating visual art. Please review the artwork image that will be presented to you and provide your evaluation.

After presenting the generated image to the Critic agent, the following prompt is provided:

Please evaluate the given image. Describe what is depicted, provide an overall evaluation with a reason, and specify any areas for improvement if necessary.

S3.2 Themes for Generation

Themes used in the experiments were devised using GPT-4o with the following prompt.

Come up with a total of 20 themes for creating an art piece, and provide them in Python list format. Each theme should be a single string of three short, common, and easy-to-understand words. The 20 themes should include:

- 4 everyday landscape scenes
- 4 natural landscapes
- 4 everyday objects
- 4 symbolic objects
- 4 poetic or abstract themes

The 20 themes (Table S1) were used. Of these, the ten odd-numbered themes were used for the Likert-scale evaluation, and the ten even-numbered themes were used for the pairwise evaluation. For each of the 20 themes, 10 images were generated using each of the 7 generation methods with different seeds, resulting in a total of 1,400 images used for evaluation.

S3.3 Measurement of Diversity

To capture qualitative rather than pixel-level variations, we projected generated images ($256 \times 256 \times 3$ pixels) into a lower-dimensional space and assessed the variance within that space. To obtain semantically meaningful feature representations, we used an image embedding model based on deep neural networks, denoted as f_{embed} . Given an image i , we extracted a feature vector $h = f_{\text{embed}}(i)$. As this vector was still high-dimensional, we further compressed it into a two-dimensional space using a dimensionality reduction function f_{reduc} , resulting in a final feature vector $h' = f_{\text{reduc}}(h)$.

Given a set of n images $i_1, i_2, \dots, i_n$ generated under a specific condition, we obtained the corresponding feature vectors $h'_1, h'_2, \dots, h'_n$ through the feature extraction and dimensionality reduction processes. Let σ_1 and σ_2 denote the variances along the first and second dimensions of these two-dimensional feature vectors, respectively. Specifically, if $h'_{k,1}$ represents the first

Table S1: List of Themes Used in the Experiments

Category	Themes
Everyday landscape scenes	1: City street corner 2: Suburban backyard view 3: Busy market square 4: Quiet park bench
Natural landscapes	5: Mountain sunrise glow 6: Ocean waves crashing 7: Forest autumn leaves 8: Desert sunset hues
Everyday objects	9: Old wooden chair 10: Vintage coffee cup 11: Rusty garden tools 12: Worn leather shoes
Symbolic objects	13: Broken hourglass time 14: Open book knowledge 15: Locked door mystery 16: Fallen feather hope
Poetic or abstract themes	17: Whispers of silence 18: Dreams in motion 19: Echoes of time 20: Shadows of memory

component of h'_k , and $\bar{h}'_1$ is its mean across the n images, then the variance is given by $\sigma_1 = \frac{1}{n} \sum_{k=1}^n (h'_{k,1} - \bar{h}'_1)^2$. σ_2 was computed similarly for the second dimension. We defined the sum of these variances, $\sigma_{\text{sum}} = \sigma_1 + \sigma_2$, as a quantitative metric of diversity, which was used in Test-Diversity.

In our experiments, we used a Vision Transformer pretrained on the ImageNet-21k dataset [3] as the embedding model f_{embed} . For dimensionality reduction, we used Principal Component Analysis (PCA). The reduction function f_{reduc} was precomputed using 2,000 images: 1,000 landscape images randomly sampled from the SUN397 dataset [9], and 1,000 images generated by Stable Diffusion, covering a diverse range of everyday scenes.

S3.4 Implementation Details

The generated images, originally sized $512 \times 512 \times 3$, were resized to $256 \times 256 \times 3$ before being output.

Stable Diffusion expects a maximum token length of 75 tokens for input prompts. Therefore, when the Artist agent generated prompts, the token length was checked and feedback was provided if the prompt exceeded 50 tokens. If no revision was observed after three iterations of feedback, the prompt was forcibly truncated to 75 tokens.

Additionally, Stable Diffusion is designed to output a black image when a Not Safe For Work (NSFW) prompt is detected. As disabling this functionality is not recommended, we conducted all experiments with NSFW filtering enabled. In cases where a black image was generated, the Artist agent was able to revise the prompt appropriately in the next loop, so this issue did not interfere with the process.

S3.5 Generated Outputs

The image generation results for each method on the themes "City street corner" and "Forest autumn leaves" are shown in Fig. S4. From these figures, it can be observed that while the outputs

of method G show everyday and realistic expressions, the other methods attempt more artistic representations. Additionally, considering that AG, AGC1, and AGC100 are methods that repeat the loop 0, 1, and 100 times, respectively, it is evident that initial ideas are progressively inherited in the order of AG, AGC1, and AGC100, when viewing the images across each seed. In the "City street corner" theme, Claude-based methods tend to adopt styles such as watercolor, monochrome, or abstract representations. In contrast, GPT-based methods more frequently use vivid colors, and in AGC100-gpt, the style returns to a relatively realistic expression. In the "Forest autumn leaves" theme, Claude-based methods consistently depict forest scenery, whereas GPT-based methods incorporate additional elements such as humans, animals, and benches.

A detailed example output is shown in Fig. S5. It illustrates how the prompt generated by the Artist is gradually refined in response to evaluations by the Critic, which in turn leads to incremental changes in the outputs from the Generator. Understanding how and to what extent changes in the prompt, generated image, and evaluation mutually influence one another remains an important topic for future analysis.

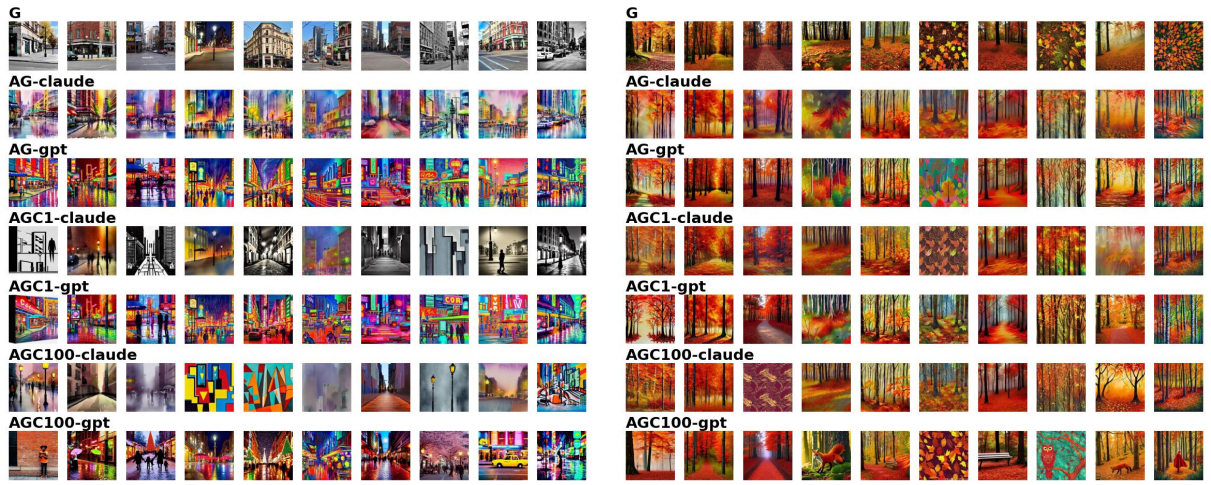


Figure S4: Generated results for the themes "City street corner" (left) and "Forest autumn leaves" (right). Each vertical column corresponds to the same seed across different generation methods. All images are AI-generated.

References

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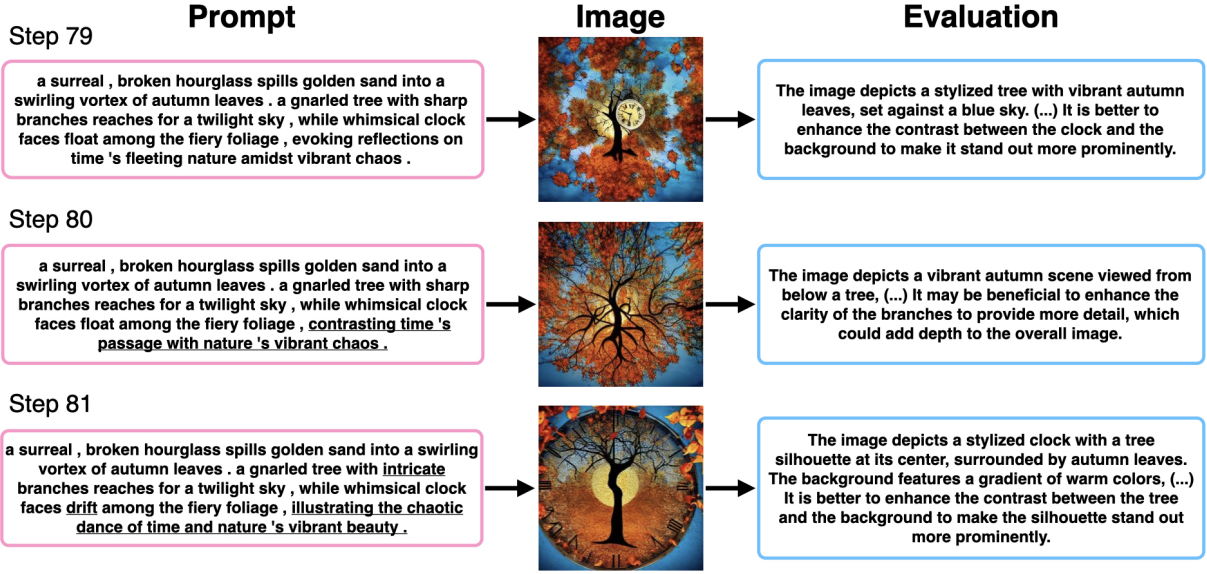


Figure S5: Example outputs from the AGC100-gpt generation method, with the theme "Broken hourglass time", using seed 9, during steps 79 to 81. For clarity, changes in the prompts are underlined and parts of the evaluation text are omitted. All images are AI-generated.

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